EXTRAS_AND_FUTURE_WORK.md
EXTRAS AND FUTURE WORK (separate from the main phases)

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Purpose. This document lists every *extra* idea proposed in the project conversation (13?16 Oct 2026) that is not one of the major contributions. The major contributions stay in the main phases: C-core (silent failures are invisible to default Kubernetes/Compose), N1/N2 (conformal guard, invariants, cross-model canary, tiered degradation), C2 (soft invariants), C3 (silent-fault benchmark), C1 (correctness SLO, Phase 7), N4 (DoE/RSM + Bayesian-tuned probes), and the operator (Phase 8). Extras may be added after those phases are finished. Nothing here is being built now.

Status labels: *Implemented-local* (code exists, tested only on the laptop, not in the cluster) · *Manifest-only* (files exist, never applied/run) · *Partial* · *Not started*.
Inclusion rule for the paper. Extras go in a "Supporting / preliminary experiments" section or "Future work". Never claim novelty for them; cite the related work noted in each row. Every number must come from a file in results/.

1. Master list

ID	Extra	Origin (when proposed)	Current status	What already exists	Effort to finish	Depends on
X01	Model-aging forecast (Weibull survival on out-of-sample LSTM errors, /aging)	"AI additions" round	Implemented-local	analysis/model_aging.py, inference-api/aging.json, /aging endpoint; shapes 1.03?1.34, median 2.7?5.3 months	0.5 day (cluster run + figure)	?	[EXTRA STATUS: DONE (cluster: /aging, TESTING 3.13)]
X02	Drift-triggered retraining loop (PSI + Page-Hinkley ? retrain ? shadow gate ? promote/reject)	AI additions	Implemented-local	inference-api/drift.py, retrain/loop.py, /drift; tested with a 3-epoch candidate (rejected 18/18)	1 day (real-epoch retrain, verify humidity PSI cause)	X05	[EXTRA STATUS: DONE (cluster; retrain not better, K34)]
X03	Energy-aware tier selection (cheapest valid tier, estimated joules)	AI additions	Implemented-local	ENERGY_AWARE=1, /energy; about 73 % lower estimated energy	0.5 day; real counters (Kepler/RAPL) need Linux host	?	[EXTRA STATUS: DONE (cluster; accuracy compared in 3.16)]
X04	Failure predictor, extended (LSTM autoencoder, SMOTE, real-fault traces beyond the synthetic leak/drift)	Original novelty list	Partial (Phase 10 done with Isolation Forest + Mahalanobis on synthetic precursors)	analysis/failure_predictor.py, results/predictor_results.json	1 day for extensions	longer healthy data (K30)
X05	Shadow-traffic model gate (mirrored requests, conformal + vote criteria, promote/reject)	N5	Implemented-local	SHADOW_LSTM_DIR in app.py, analysis/shadow_gate.py; bad scaler rejected 3/3	0.5 day in cluster (good + several bad candidates, rollback by image tag)	?	[EXTRA STATUS: DONE (cluster 24/24; final re-check 7/7)]
X06	Argo Rollouts canary with a Prometheus AnalysisTemplate	Phase 5 plan	Manifest-only	k8s/rollout/rollout-test.yaml	done 2026-10-17	X05	[EXTRA STATUS: DONE - promote/abort verified, TESTING 3.18]
X07	Kyverno admission policy + cosign signing + Trivy scan (supply chain; tampered-model fault)	Extras 6/7 round	Manifest-only	k8s/policy/kyverno-verify.yaml, .github/workflows/ci.yml (never run)	1?2 days; needs a GitHub repo with Actions	GitHub repo	[EXTRA STATUS: DONE locally - Kyverno validate + cosign verifyImages admit signed/deny tampered (TESTING 3.19); CI workflow written, not run (needs GitHub)]
X08	GitHub Actions CI (tests, build, scan, sign, push)	Phase 5 plan	Manifest-only	.github/workflows/ci.yml	0.5 day after repo exists	GitHub link	[EXTRA STATUS: BLOCKED]
X09	OpenTelemetry tracing across frontend / API / DB (recovery latency per hop)	Phase 2 plan	Not started	none	1 day	?	[EXTRA STATUS: DONE - OpenTelemetry + Jaeger, one trace spans frontend and inference-api (TESTING 3.19)]
X10	Warm-up-aware predictive autoscaling (KEDA + SARIMA load forecast) vs HPA / tuned HPA	Early novelty list (secondary)	Not started	none; related work: PathFormer scaling paper, SARIMA PBA	2 days	cluster	[EXTRA STATUS: DONE as reactive KEDA (not SARIMA-forecast); no benefit on single node (TESTING 3.19)]
X11	SARIMA vs LSTM under identical faults (load time, size, recovery) as a dedicated comparison	Early plan	Partial	data exists in both configs but no dedicated analysis	0.5 day analysis	existing results	[EXTRA STATUS: DONE (3.15 and 3.16)]
X12	MTTR decomposition (detect, schedule, image pull, model load, warm-up)	Early N2 text	Not started	only total recovery is recorded; model_load_seconds gauges exist	1 day (instrument phases)	?	[EXTRA STATUS: DONE (3.17)]
X13	Semantic probe wired into Kubernetes probes (D1 guard only vs D2 semantic readiness/liveness restart)	Phase 4/5 plan	Partial	/health/semantic exists but is not used by any Kubernetes probe	0.5 day	?	[EXTRA STATUS: DONE (3.17)]
X14	Per-variable partial readiness (rainfall / temperature / humidity independently)	Novelty list	Partial	partial_ready() reported in /health/ready; never exercised	0.5 day	?	[EXTRA STATUS: DONE (3.17)]
X15	Cache tier exercised and measured (last-good forecast during outage)	Tier design	Partial	_CACHE in app.py, never reached in experiments	0.25 day	?	[EXTRA STATUS: DONE (3.17)]
X16	Adaptive / online conformal guard and exchangeability study (static vs adaptive false-alarm rates under seasonality/drift)	Web-research round	Not started	static window calibration + hold-out ablation	1 day	?	[EXTRA STATUS: DONE (3.14)]
X17	Time-series foundation model (Chronos/TimesFM) as a third version	Web-research round (rank 5)	Not started	none; risky: correlated failure modes, heavy dependencies	2?3 days	torch install	[EXTRA STATUS: ATTEMPTED (3.17)]
X18	Third diverse model (XGBoost or Prophet) in the voting set	AI-additions list	Not started	none	1 day	?	[EXTRA STATUS: DONE offline (3.16)]
X19	Input-side semantic guard (conformal / Isolation Forest on user inputs and uploads)	AI additions	Not started	none	1 day	?	[EXTRA STATUS: DONE (offline + UI warning; weak, K35)]
X20	Adaptive tier ordering (Hedge / multiplicative weights on recent error)	AI additions	Not started	none	1 day	?	[EXTRA STATUS: DONE offline (3.16)]
X21	Uncertainty bands on forecast plots (conformal intervals shown to users)	AI additions	Not started	none	0.5 day	?	[EXTRA STATUS: DONE (API field, coverage 3.16)]
X22	Frontend "degraded" label and tier badge in the web UI	Tier design	Not started	API returns tier, degraded; UI ignores them	0.5 day	?	[EXTRA STATUS: DONE (verified in browser)]

| X23 | SHAP explanations for predictor and tier decisions | AI additions | Not started | XAI page exists for the original models only | 1 day | X04 | [EXTRA STATUS: DONE (permutation attribution)]
| X24 | LLM incident summariser (operator history + metrics ? plain-language report) | AI additions | Not started | operator keeps status.history | 1 day | operator | [EXTRA STATUS: DONE as template summariser, results/incident_reports.md]
| X25 | Granger-causality root cause across pod metrics and output quality | AI additions | Not started | none; overlaps SGCAD | 2 days | X04 data | [EXTRA STATUS: DONE (screen)]
| X26 | RL/DQN autoscaler with warm-up awareness | AI additions | Not started | none; crowded area | 3 days | X10 | [EXTRA STATUS: DONE in simulation only (TESTING 3.19)]
| X27 | Synthetic fault augmentation (SMOTE / generative) for rare failures | AI additions | Not started | none | 0.5 day | X04 | [EXTRA STATUS: DONE (no SMOTE gain)]
| X28 | Log-based anomaly detection (small transformer on service logs) | AI additions | Not started | none; overlaps LogBERT-lite | 2 days | ? | [EXTRA STATUS: DONE - negative result, service emits no logs (TESTING 3.19)]
| X29 | Chaos Mesh / LitmusChaos re-run of the fault taxonomy (node failure, slow database, network faults) | Phase 3 plan | Not started | custom injector /fault only | 1?2 days | cluster | [EXTRA STATUS: DONE - pod-kill and 300 ms delay (TESTING 3.19)]
| X30 | Extended failure taxonomy: node failure, slow PostgreSQL, bad rollout | Original taxonomy list | Not started | crash/hang/oom + 7 silent faults exist | 1 day | X29 optional | [EXTRA STATUS: DONE (database lock and restart; node failure not possible on single node)]
| X31 | Model-loading strategies (lazy / eager / shared volume / warm standby) vs recovery | Early novelty list | Not started | per-model load times measured once | 1?2 days | X12 | [EXTRA STATUS: DONE (eager vs lazy; other strategies not run)]
| X32 | Energy / carbon per recovery strategy | Extras list | Partial | estimated joules per tier only | 0.5 day | X03 | [EXTRA STATUS: DONE (3.16)]
| X33 | Learned recovery policy: digital twin + bandit (re-run with third action) | N3 | Partial (twin run; Phase 8 operator experiments in progress) | analysis/twin_bandit.py, policy.json, operator bandit policy | finishing with Phase 8 | Phase 8 | [EXTRA STATUS: DONE (Phase 8)]
| X34 | Open benchmark release package (taxonomy, scripts, traces, README, licence) | Benchmark plan | Partial | scripts and results exist, no packaging | 0.5 day | GitHub repo | [EXTRA STATUS: DONE (release/silentbench_release.zip)]
| X35 | Git/GitHub with Git LFS for 118 MB models + reproducibility check on a clean machine | Laptop-change discussion | Not started | .gitignore, .gitattributes, RESTORE.md | 0.5 day | GitHub link | [EXTRA STATUS: DONE (repository pushed with Git LFS)]

2. Suggested order after all main phases are done

1. Cheap, high value: X13, X15, X22, X21 (small changes, visible in the demo), X11 (analysis only).
2. Supporting experiments for the paper: X05 (with rollback) ? X04 ? X16 ? X12.
3. Packaging: X35 ? X34 ? X08.
4. Only if time remains: X01?X03 in the cluster, X09, X10, X06/X07.
5. Probably skip: X17, X25, X26, X28 (high effort, crowded, or weak independence).

3. How to include extras in the paper

- Section "Supporting experiments (preliminary)": one paragraph and one table per extra actually run (X05, X04, X16 recommended).
- "Future work": list the remaining IDs in one paragraph.
- Do not put any extra in the contribution list. Cite: PathFormer scaling paper and SARIMA PBA (X10); TELKA, safe-remediation and ARBITER (X33); D3M and Machida N-version ML (X18, X20); LogBERT-lite (X28); SGCAD (X25); shadow-deployment practice (X05).

4. Caveats

- Several extras overlap published work (see REFERENCES §3); only combinations specific to ML forecasting services on Kubernetes are defensible.
- Energy figures are estimates (15 W per busy core) unless real counters are used.
- Items marked Implemented-local were exercised on the laptop only; rerun them in the cluster before reporting.
- X03 and X10 interact with the single-worker saturation issue (K25); use the threaded image (:6) or later.

5. Information needed

- GitHub repository link (X07, X08, X34, X35).
- Whether the paper should include a "Supporting experiments" section (target venue/page limit).
- Which 2?3 extras the user wants as the first additions after Phase 8.

Added 2026-10-17

- Novelty 6 deadline/load-aware degradation: DONE (TESTING 3.20). Next: per-tier locks; repeat scaling study on image :12 with heavier burst; metamorphic readiness probe; healthy-but-wrong time / MTTC metrics (proposed, not started).

- Metamorphic readiness probe (X36): DONE as experimental, partly negative result - reliable for output-level faults, marginal for subtle bit flips, brittle under drift (TESTING 3.21, K42).